EchoForge and Engineered Intelligence for Low-False-Alarm UAS Radar: A Strict-Open Synthetic Benchmark

Jeppson Taylor, Lead Author
NeverHuman Research Group

***Abstract*—Low-altitude uncrewed aerial system (UAS) radar detection is a rare-positive, false-alarm-constrained problem. EchoForge is a strict-open, radar-first synthetic sensing benchmark for fixed-wing pusher-prop public-proxy UAS research. It exposes public-source object priors, radar branch cards, synthetic complex IQ, range-Doppler products, acoustic and passive-RF cues, detector-view contracts, group-locked splits, and reproducible evidence without claiming measured-platform truth, proprietary-equivalent behavior, classified fidelity, or deployment performance. The benchmark run contains 10000 scenario groups, 50 positive groups, and 30000 phase records. Candidate discovery, calibration, thresholding, and model selection use train/CV rows only; the 4500-record blind holdout is scored once after lock. The accepted comparator is the strongest human-engineered sensor-fusion lane assembled from the radar and cue branches. Engineered Intelligence (EI) improves point Recall@≤1%FPR from 0.292 to 0.833 and improves LCB95 Recall@≤1%FPR from 0.083 to 0.713. It raises AP from 0.128 to 0.830 and cuts selected-threshold false positives from 49 to 3. EI gives up global ROC AUC relative to prior fusion, 0.916 vs. 0.938, so the claim is precise: EI improves the low-FPR operating point on this declared synthetic public-proxy benchmark, not universal ranking dominance or operational performance.**

***Index Terms*—Synthetic radar, UAS detection, radar simulation, micro-Doppler, sensor fusion, engineered intelligence, low false alarm detection, reproducibility.**

I. Introduction

Radar detection for low-altitude UAS is difficult because the target is small, intermittent, and often embedded in clutter or hard-negative traffic that produces plausible false tracks. In this setting, ROC AUC is not enough. A detector can rank positives well on average and still fail once the false-positive budget is nearly exhausted. EchoForge therefore treats LCB95 Recall@≤1%FPR as the claim-bearing metric, with AP, ROC AUC, calibration, selected-threshold confusion counts, and false-positive families as guardrails.

This paper has one experimental spine: simulator assumptions, human detector branches, accepted human fusion, EI search, selection lock, and one blind holdout pass. The paper-facing positive class is a fixed-wing pusher-prop public proxy. The proxy is inspired by open public descriptions of fixed-wing pusher-prop UAS families, but the benchmark does not claim measured Iranian-platform radar signatures, route behavior, payload behavior, fielded-system equivalence, or classified fidelity.

The contributions are:

- a strict-open synthetic radar and multimodal sensing benchmark for fixed-wing pusher-prop public-proxy UAS scenarios;
- a radar-first evaluation setup connecting waveform assumptions, range-Doppler structure, CFAR/MTD scoring, micro-Doppler cues, tracking, and fusion;
- a human-method ladder that ends in accepted human sensor fusion before EI is introduced;
- an auditable EI method that evolves sparse calibrated fusion on train/CV evidence and is judged against the accepted comparator under one group-locked protocol;
- a focused figure set: simulator stack, phase ladder, EI train/CV evolution, false-positive burden, and qualitative range-Doppler examples.

II. Radar Processing Background

The radar branch begins with a received-power proxy,

$$P_r \propto \frac{P_t G_t G_r \lambda^2 \sigma(\theta, f, \phi)}{(4\pi)^3 R^4 L}, \tag{1}$$

where σ is an aspect- and frequency-dependent public-proxy RCS term, R is slant range, and L is aggregate loss. EchoForge then synthesizes complex IQ as a target return plus clutter, interference, and receiver impairment,

$$x_{m,n} = a_{m,n} e^{j(2\pi f_d m T_r + \varphi_n)} + c_{m,n} + \epsilon_{m,n}. \tag{2}$$

The range-Doppler budget follows standard radar processing: bandwidth controls range resolution, CPI controls Doppler resolution, and PRF constrains ambiguity. The derived quantities used in the branch cards are

$$\Delta R = \frac{c}{2B}, \qquad \Delta f_D = \frac{1}{T_{\text{CPI}}}, \qquad \Delta v = \frac{\lambda}{2T_{\text{CPI}}}. \tag{3}$$

Classical detector families then ask whether a cell, Doppler ridge, or short track is locally exceptional. CFAR compares a peak against local clutter; MTD emphasizes coherent motion; track scoring rewards persistence; micro-Doppler features summarize modulation from rotating or flapping structures; and fusion combines sensor-local evidence under uncertainty [1]–[7]. Public measured radar datasets and UAS micro-Doppler studies are useful as observable-family anchors, but EchoForge uses them as compare-only references rather than as proof of measured positive-class truth [8]–[13].

TABLE I
RADAR BRANCH CARDS USED IN THE BENCHMARK

Branch	f_0 GHz	BW MHz	CPI ms	ΔR m
X/Ku C-UAS	10.0	600	6.0	0.250
S-band surveillance	3.1	180	6.0	0.833
GBAD cueing	10.3	300	6.0	0.500

III. EchoForge Simulator

EchoForge is the simulator and evidence stack behind the experiment. It separates public-proxy priors, scenario construction, radar and cue synthesis, detector-view materialization, split-aware evaluation, and paper evidence. Figure 1 is the paper-facing map of that stack.

The simulator exposes six capabilities needed for this benchmark:

- *Public-proxy object priors*: fixed-wing pusher-prop geometry, broad kinematics, aspect/RCS uncertainty, pusher-prop cadence, phase labels, and hard-negative families.
- *Radar branch cards*: X/Ku C-UAS, S-band surveillance, and GBAD cueing archetypes with explicit carrier, bandwidth, CPI, PRF/CRF, chirp count, range resolution, Doppler resolution, and velocity resolution.
- *Scene and receiver stress*: non-Gaussian clutter, weather/terrain glint, vegetation motion, RFI bursts, multipath ghosts, AGC compression, drift, quantization, dropped CPI, Doppler folding, and calibration offset.
- *Multimodal cue streams*: acoustic cadence/agreement and passive-RF absence, contamination, sparse cue geometry, and provenance quality.
- *Detector-view contracts*: model-facing features include detector scores and cue summaries; labels, split fields, group IDs, audit headers, and generator internals are denied to the model.
- *Reproducible evidence*: deterministic seeds, group-locked splits, strict train/CV selection, one blind-holdout score pass, and generated figure/code appendices.

IV. Benchmark Design

The benchmark run contains 10000 scenario groups, 50 positive groups, and 30000 phase records. Each group expands into three fixed windows: initial take-up (0–30 s), climb transition (30–90 s), and cruise altitude (90–150 s). Split assignment occurs at the scenario-group level before phase expansion; all phase rows from the same group inherit the same split. The blind holdout contains 4500 records, including 24 positive records from 8 positive scenario groups.

The primary KPI is LCB95 Recall@≤1%FPR. It is the lower 95% group-block bootstrap bound for recall at FPR $\leq 1\%$. The point estimate Recall@≤1%FPR is also reported, but LCB95 Recall@≤1%FPR is claim-bearing because the positive holdout is group-locked and small. Selected-threshold TP/FP/FN counts are kept separate from swept fixed-FPR metrics.

V. Human Detection Ladder

The human ladder is ordered from sensor-local evidence to accepted sensor fusion. Every method uses the same three phase windows and train/CV-only threshold or calibration basis. Figure 2 repeats that method order across the three phases.

VI. Engineered Intelligence

Human radar engineering is powerful, but it has three bottlenecks: knowledge consumption, idea generation, and validation bandwidth. EI is the engineering response to those bottlenecks. It is not an opaque model handle. In this benchmark it is controlled algorithm discovery: candidate score families are generated, sparse combinations are ranked on train/CV evidence, a monotone calibrator is fit, a selection lock is written, and the blind holdout is scored once.

A. Method Math

Let x_i be the detector-view feature vector for record i. Candidate component j maps that view to a bounded score

$$s_{ij} = g_j(x_i). \tag{4}$$

EI then searches sparse nonnegative late fusion weights,

$$z_i = \sum_{j=1}^{k} w_j s_{ij}, \qquad w_j \geq 0, \qquad \sum_{j=1}^{k} w_j = 1. \tag{5}$$

The selected fused score is calibrated through monotone geodesic odds,

$$\hat{p}_i = \sigma\left(a \log \frac{z_i + \epsilon}{1 - z_i + \epsilon} + b\right). \tag{6}$$

The train/CV objective follows the implementation: it rewards AP and ROC AUC while penalizing excess FPR and weak initial-take-up behavior,

$$\begin{aligned} J = {} & \mathrm{AP}_{\mathrm{CV}} + 0.25\,\mathrm{AUC}_{\mathrm{CV}} \\ & - \alpha\,\max(0, \mathrm{FPR}_{\mathrm{CV}} - \tau) - \beta\,\Delta_{\mathrm{takeup}}. \end{aligned} \tag{7}$$

All terms are computed on train/CV rows. Holdout rows are not used to choose candidates, calibrators, thresholds, components, or weights.

VII. Results

The gain is concentrated where the benchmark is designed to be hard: low-FPR recall. Table III is the headline ledger and separates swept low-FPR recall from selected-threshold confusion counts and ranking guardrails.

The correct interpretation is improved operating behavior under a strict low-FPR cap. EI does not dominate every metric: accepted human fusion retains the higher global ROC AUC. That trade-off is why the paper reports AP, ROC AUC, confusion counts, FPR, F1, and false-positive family burden instead of claiming universal ranking dominance.

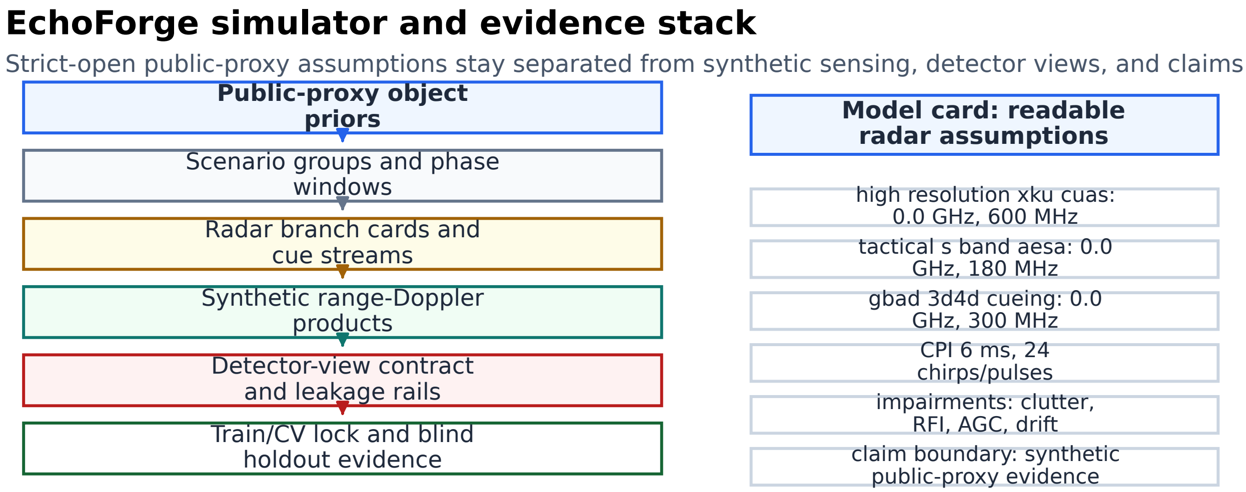

Fig. 1. EchoForge simulator and evidence stack. Public-proxy object priors, scenario groups, radar branch cards, synthetic range-Doppler products, detector-view contracts, train/CV locks, blind holdout evidence, and claim-boundary language are kept separate so the benchmark can be audited without conflating synthetic products with measured truth.

VIII. False-Positive Burden

False positives are the most intuitive failure mode in a low-FPR radar paper. Figure 4 shows which hard-negative families consume the selected-threshold false-alarm budget for each method.

The family plot highlights the central operational difference: accepted human fusion spends many more selected-threshold false alarms across bird, RC fixed-wing, weather, multipath, terrain/glint, and rotating-structure stressors. EI leaves only 3 selected-threshold false positives on the same holdout. This is robustness evidence for the declared synthetic benchmark, not an evasion or deployment claim.

IX. Why It Matters

The result matters because it changes the detector-development loop. Classical radar work still matters: without plausible waveform assumptions, detector views, split hygiene, calibration, and accepted fusion, there is no trustworthy benchmark. EI does not replace that foundation. It compounds it.

Once the problem is decomposed into auditable stages, the system can search for transforms, combinations, and calibrators that a human team might not propose first. The final artifact remains code, weights, math, and evidence. The invention process becomes continuous. Progress is then limited less by a fixed menu of manually enumerated ideas and more by simulator quality, evidence quality, compute, search discipline, and holdout hygiene.

X. Limitations

The corpus is synthetic, single-seed, and public-proxy. It has 50 positive groups total and 8 positive groups in the holdout split. Group-block intervals are necessary but not enough for broad claims. EchoForge does not provide measured Iranian-drone radar truth, fielded-sensor equivalence, hardware-in-the-loop validation, track-level operational FAR, route/evasion modeling, or named-platform performance. The ROC AUC guardrail also matters: accepted human fusion ranks better globally, while EI performs better at the low-FPR operating point chosen as the benchmark KPI.

The next evidence steps are multi-seed corpora, larger positive holdouts, richer weather/RFI/site stressors, measured compare-only anchors expanded by sensor branch, hardware-in-the-loop hooks, and a formal pre-registration file for every promoted EI search.

XI. Conclusion

EchoForge provides a strict-open radar simulation and evidence stack for public-proxy UAS detection: explicit assumptions, radar branch cards, hard negatives, detector-view denylisting, group-locked splits, train/CV-only selection, and reproducible evidence. The paper's central result is simple: accepted human sensor fusion is strong, but EI-discovered sparse calibrated fusion is much stronger at the low-FPR operating point on this benchmark. EI raises LCB95 Recall@$\leq$1%FPR from 0.083 to 0.713 and cuts selected-threshold false positives from 49 to 3. The future implied by this result is not a one-off model; it is a radar-development system in which novel math and code evolve in response to data under strict evidence controls.

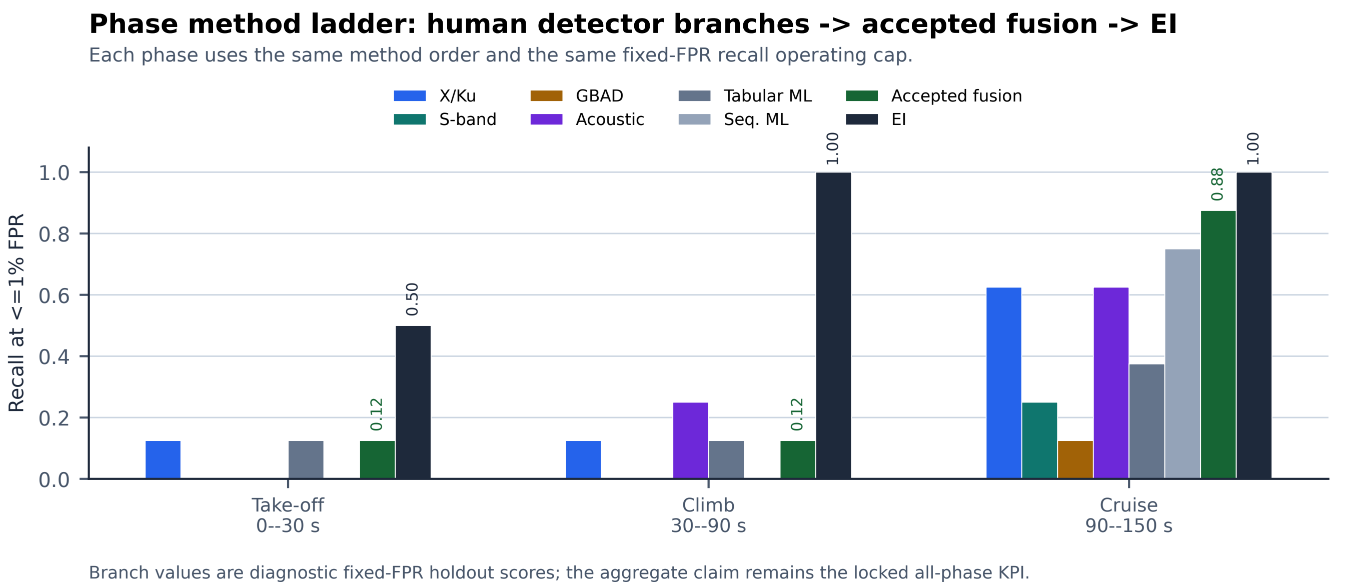

Fig. 2. Human detection ladder and EI challenger by phase. The same take-off, climb, and cruise panels show progression from radar and cue branches to ordinary ML controls, accepted human fusion, and EI. The plotted diagnostic is Recall@$\leq$1%FPR on the holdout phase slice; phase conclusions remain diagnostic because each phase has only eight positive holdout records.

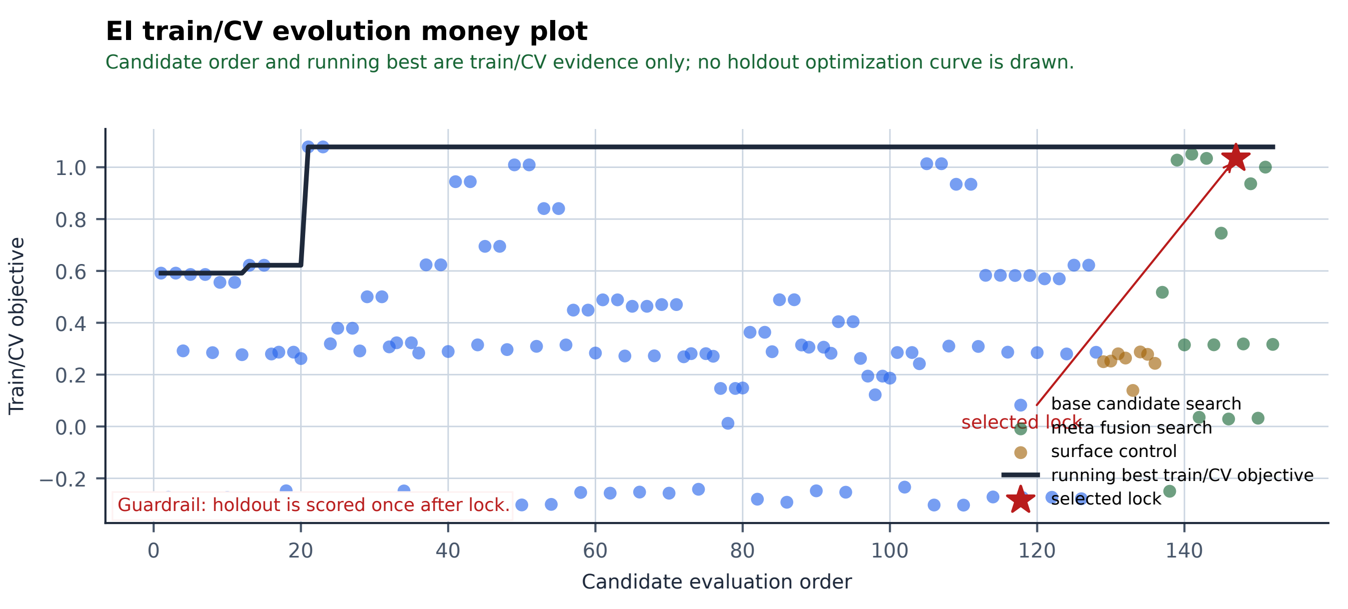

Fig. 3. EI train/CV evolution money plot. Candidate evaluations are shown in actual train/CV order with the running best emphasized and the selected lock marked. The blind holdout is scored once after lock and is not used to draw this curve; there is no holdout optimization curve.

TABLE II
METHOD LADDER BEFORE AND INCLUDING EI

Method	Sensors used	Phase windows	Processing pipeline	Threshold/calibration basis	Role
X/Ku radar	X/Ku radar	0–30, 30–90, 90–150 s	CFAR-like power contrast, range-Doppler concentration, micro-Doppler spread	train/CV max-F1	high-resolution radar sanity branch
S-band radar	S-band radar	same	MTD/range-velocity summaries and track confidence	train/CV max-F1	lower-resolution surveillance stress check
GBAD cueing	GBAD cue radar	same	horizon, revisit, stale-track, and cue-confidence terms	train/CV max-F1	cueing branch rather than final classification
Acoustic cue	acoustic nodes	same	cadence, stability, and node agreement	train/CV score threshold	non-radar propulsion-cadence support
Passive RF	passive-RF provenance	same	RF silence, RFI burst, sparse cue geometry, and provenance quality	train/CV score threshold	missingness/provenance stress
Tabular/sequence ML	denylisted detector views	same	ordinary tabular and sequence controls using detector-facing features only	train/CV max-F1	non-EI learned controls
Accepted fusion	radar, acoustic, passive-RF context	same	human late fusion with source confidence, cross-sensor confirmation, and stale-track penalties	train/CV calibrated threshold	main non-EI comparator
EI sparse fusion	all detector views plus evolved components	same	sparse nonnegative late fusion with monotone odds calibration	EI selection lock from train/CV	locked challenger

TABLE III
BLIND-HOLDOUT RESULT VERSUS ACCEPTED HUMAN FUSION

Metric	Accepted human fusion	EI sparse fusion	Change
LCB95 Recall@$\leq$1%FPR	0.083	**0.713**	**+0.630; +760.0%; 7.60x**
Point Recall@$\leq$1%FPR	0.292	**0.833**	**+0.542; +185.7%; 1.86x**
Average precision	0.128	**0.830**	**+0.702; +547.4%**
Selected-threshold TP/FP/FN	10/49/14	**19/3/5**	**46 fewer FP**
Selected-threshold FPR	0.010947	**0.000670**	**93.9% fewer false alarms**
F1	0.241	**0.826**	**+0.585**
ROC AUC	**0.938**	0.916	**-0.021 guardrail**

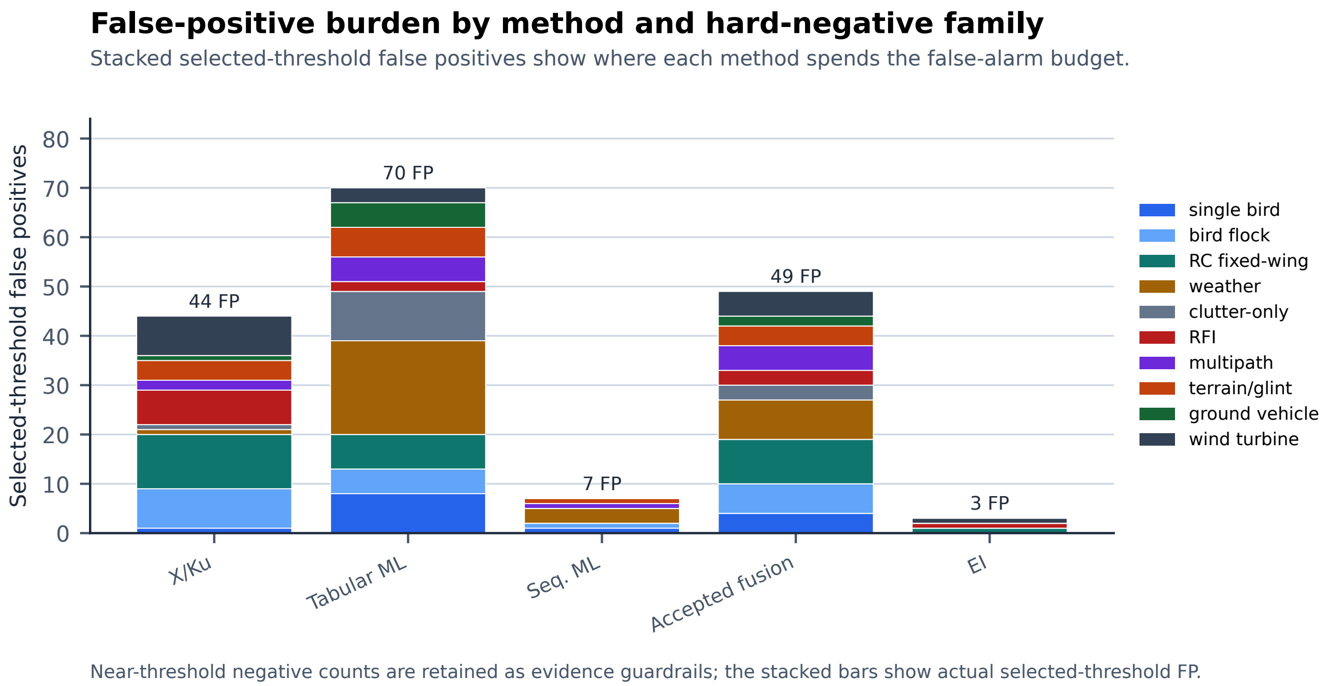

Fig. 4. Selected-threshold false-positive burden by method and hard-negative family. Bird includes single-bird and flock-like confusers; RC fixed-wing includes small model aircraft with prop-cadence overlap; weather includes rain, dust, and weather-cell stress; clutter-only includes no-target scene clutter; RFI includes burst contamination and passive-RF artifacts; multipath includes ghost/reflection-like tracks; terrain/glint includes site reflections; ground vehicle includes low-altitude track-like vehicle confusers; wind turbine is a rotating-structure hard negative. Near-threshold negatives are retained as evidence guardrails rather than stacked as false positives.

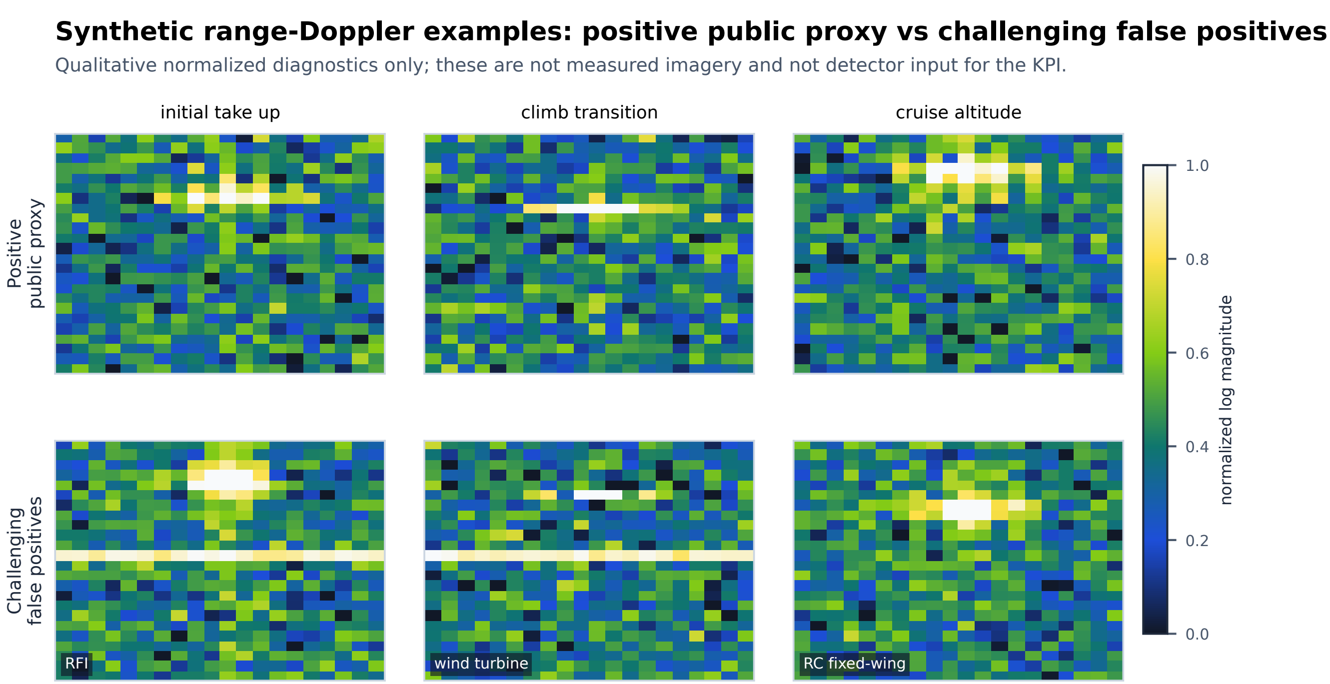

Fig. 5. Qualitative synthetic range-Doppler examples: fixed-wing pusher-prop public-proxy positives versus challenging EI false positives. Axes are normalized range-Doppler bins and color is normalized log magnitude. These panels are qualitative diagnostics only, not measured imagery and not detector input for the KPI.

TABLE IV
PUBLIC-PROXY MODELING BOUNDARY

Family	Used as	Supports	Does not support
Positive public proxy	fixed-wing pusher-prop geometry, broad kinematics, aspect/RCS uncertainty, pusher cadence	synthetic detector comparison and phase stress	measured Iranian-platform signature or operational performance
Radar branches	public role envelopes for X/Ku, S-band, and GBAD cueing	branch-level observable assumptions and resolution budget	vendor sensitivity, ECCM, exact Pd/Pfa, or military performance
Hard negatives	bird, RC fixed-wing, weather, clutter-only, RFI, multipath, terrain/glint, ground vehicle, wind turbine	robustness stress and false-positive analysis	evasion optimization or field-rate estimates
Measured anchors	compare-only public radar observable families	distribution-shape sanity checks	replacement for positive-class validation

APPENDIX A
MODELING BOUNDARY APPENDIX

The fixed-wing pusher-prop public proxy uses public-source geometry, broad kinematics, aspect/RCS uncertainty, pusher-prop cadence, and phase-window assumptions. Those assumptions support synthetic detector comparison only. They do not establish measured positive-class radar truth, named-platform equivalence, proprietary sensor behavior, route behavior, payload inference, or field performance.

APPENDIX B
REPRODUCIBILITY CLI APPENDIX

These commands reproduce the paper lane. They are benchmark commands, not field-deployment instructions.

Generate the synthetic scenario and sensor corpus

```
rtk python3 -m detection.generate_main_run \
  --profile fixed-wing-pusher-proxy-v2 \
  --out-root outputs/training-data/runit-fixed-wing-pusher-proxy-v2-main-run \
  --scenario-groups 10000 \
  --seed 202605210136 \
  --force
```

Run human detector and accepted-fusion baselines

```
rtk python3 -m detection.run_main_run_detectors \
  --data-root outputs/training-data/runit-fixed-wing-pusher-proxy-v2-main-run \
  --out-root outputs/detection/runit-fixed-wing-pusher-proxy-v2-main-run \
  --folds 5 \
  --seed 202605210136 \
  --force
```

Run EI search, lock, and one-pass holdout scoring

```
rtk python3 -m detection.run_advanced_main_run_detectors \
  --data-root outputs/training-data/runit-fixed-wing-pusher-proxy-v2-main-run \
  --out-root outputs/detection/runit-fixed-wing-pusher-proxy-v2-main-run-advanced-evolution \
  --folds 5 \
  --seed 202605210136 \
  --search-profile v2_aggressive \
  --candidate-limit 128 \
  --evolution-rounds 5 \
  --evolution-sample-rows 4500 \
  --write-component-scores \
  --force
```

Build evidence, macros, figures, appendix, and PDF

```
rtk python3 -m detection.paper_evidence --strict-ei-trace --force
rtk python3 paper/generate_metric_macros.py --strict
rtk python3 paper/generate_source_appendix.py --strict
rtk python3 paper/generate_figures_focused.py --strict
rtk bash paper/build.sh --copy-tracked
```

Appendix C
High-Resolution Radar Sample Appendix

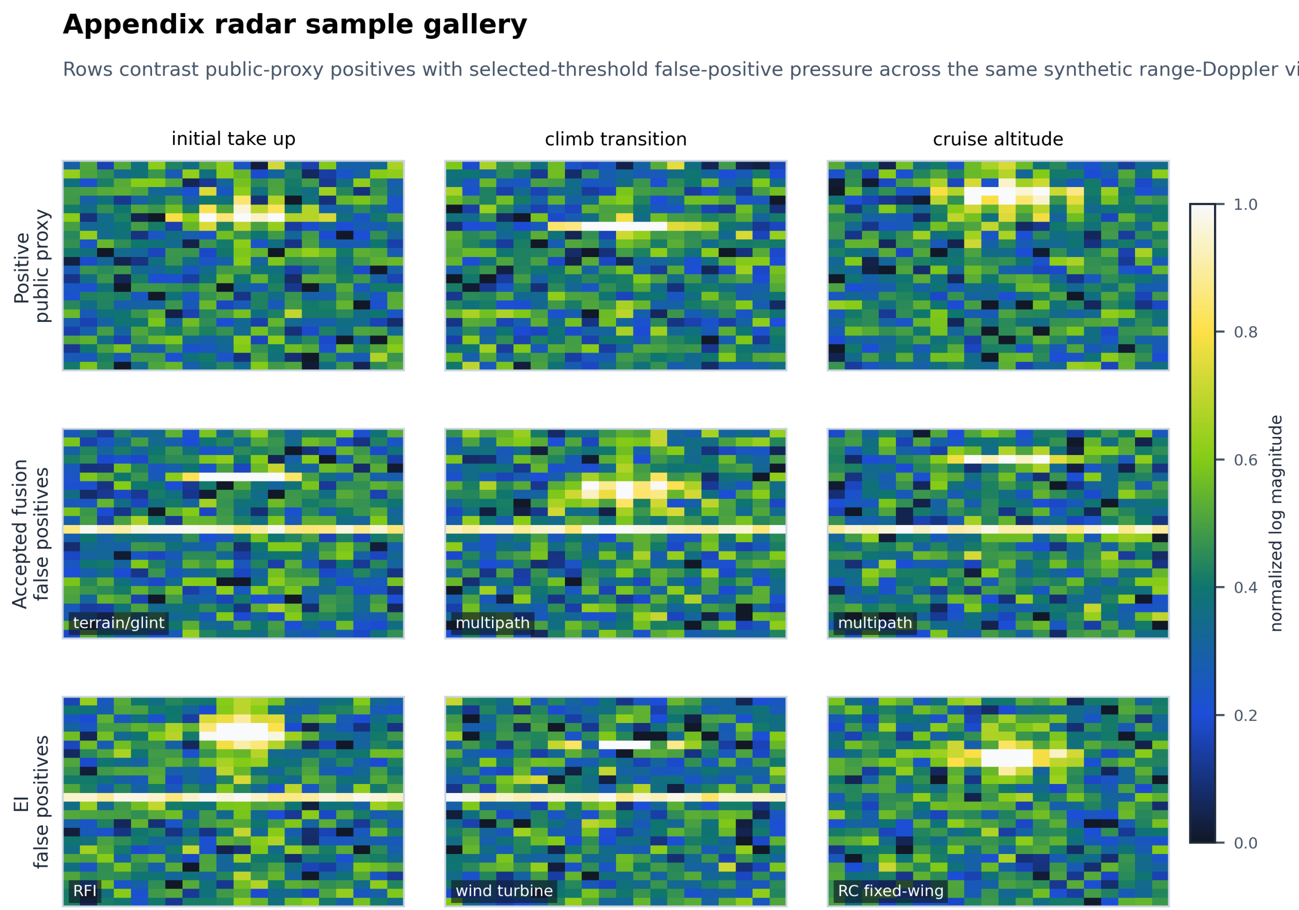

Fig. 6. Appendix radar sample gallery. Rows contrast public-proxy positives with accepted-fusion and EI selected-threshold false-positive pressure across the same synthetic range-Doppler view. These normalized diagnostics are not measured imagery and are not detector input for the KPI.

Appendix D
Runtime Code Appendix

This appendix shows only compact runtime math. Blue blocks are human-origin baseline code, green blocks are generated/evolved EI code, and lavender blocks are mixed-origin glue. Repository paths, file writers, paper plumbing, and CLI wrappers are omitted from the rendered paper.

A. Human Radar Score Routines

robust_scale

```
def robust_scale(values: np.ndarray) -> np.ndarray:
    """Scale a detector score with median/MAD robustness."""
    median = float(np.median(values))
    mad = float(np.median(np.abs(values - median)) + 1.0e-6)
    return np.clip((values - median) / mad, -12.0, 12.0)
```

cfar_range_doppler_score

```
def cfar_range_doppler_score(iq: np.ndarray) -> float:
    """Human radar branch: CFAR-like contrast in range-Doppler power."""
    power = np.abs(np.fft.fftshift(np.fft.fft2(iq))) ** 2
    clutter_median = float(np.median(power))
    clutter_mad = float(np.median(np.abs(power - clutter_median)) + 1.0e-6)
    return float((np.max(power) - clutter_median) / clutter_mad)
```

mtd_doppler_score

```
def mtd_doppler_score(iq: np.ndarray) -> float:
    """Human radar branch: moving-target Doppler concentration."""
    doppler_power = np.abs(np.fft.fft(iq, axis=0)) ** 2
    return float(np.max(doppler_power) / (np.mean(doppler_power) + 1.0e-6))
```

track_continuity_score

```
def track_continuity_score(iq: np.ndarray) -> float:
    """Human cueing branch: range contrast discounted by pulse instability."""
    power = np.abs(iq) ** 2
    range_power = power.mean(axis=0)
    pulse_power = power.mean(axis=1)
    range_contrast = float(np.max(range_power) / (np.mean(range_power) + 1.0e-6))
    pulse_continuity = float(1.0 / (1.0 + np.std(np.diff(pulse_power))))
    return range_contrast * pulse_continuity
```

B. Human Cue Score Routines

distributed_acoustic_score

```
def distributed_acoustic_score(acoustic: np.ndarray) -> float:
    """Human cueing branch: propulsion cadence and node agreement."""
    spectrum = np.abs(np.fft.rfft(acoustic, axis=1))
    peak = np.max(spectrum[:, 1:], axis=1)
    mean = np.mean(spectrum[:, 1:], axis=1) + 1.0e-6
    peak_ratio = peak / mean
    node_peaks = np.argmax(spectrum[:, 1:], axis=1).astype(np.float64)
    agreement = 1.0 / (1.0 + np.std(node_peaks))
    return float(np.mean(peak_ratio) * agreement)
```

passive_rf_provenance_score

```
def passive_rf_provenance_score(cue_quality: float, rfi_burst: float, no_signal: float) -> float:
    """Human cueing branch: sparse passive-RF provenance, not emitter identity."""
    clean_quality = np.clip(cue_quality, 0.0, 1.0)
    contamination = np.clip(0.65 * rfi_burst + 0.35 * no_signal, 0.0, 1.0)
    return float(np.clip(clean_quality * (1.0 - contamination), 0.0, 1.0))
```

C. Accepted Human Fusion

accepted_prior_fusion

```
def accepted_prior_fusion(features: np.ndarray, labels: np.ndarray) -> CalibratedFusion:
    """Human accepted fusion: train/CV-only Gaussian log-odds weights."""
    labels = labels.astype(np.int8)
    positive = features[labels == 1]
    negative = features[labels == 0]
    if len(positive) == 0 or len(negative) == 0:
        return CalibratedFusion(np.ones(features.shape[1], dtype=np.float64) * 0.1, 0.0)
```

```python

    positive_mean = np.mean(positive, axis=0)
    negative_mean = np.mean(negative, axis=0)
    pooled_var = np.var(features, axis=0) + 1.0e-3
    weights = (positive_mean - negative_mean) / pooled_var

    prior = (float(len(positive)) + 0.5) / (float(len(labels)) + 1.0)
    midpoint = 0.5 * (positive_mean + negative_mean)
    intercept = math.log(prior / (1.0 - prior)) - float(np.dot(weights, midpoint))
    return CalibratedFusion(weights=weights, intercept=intercept)
```

D. EI Sparse Fusion and Geodesic-Odds Calibration

geodesic_odds_calibration (human-origin)

```python
def geodesic_odds_calibration(score: np.ndarray, labels: np.ndarray) -> tuple[float, float]:
    """Mixed-origin EI calibration: monotone logit-on-logit fit."""
```

geodesic_odds_calibration (mixed-origin)

```python
    eps = 1.0e-6
    bounded = np.clip(score, eps, 1.0 - eps)
    geodesic_odds = np.log(bounded / (1.0 - bounded))
    y = labels.astype(np.float64)
    x_mean = float(np.mean(geodesic_odds))
    y_mean = float(np.mean(y))
    slope = float(
        np.dot(geodesic_odds - x_mean, y - y_mean)
        / (np.dot(geodesic_odds - x_mean, geodesic_odds - x_mean) + eps)
    )
    intercept = y_mean - slope * x_mean
    return slope, intercept
```

ei_sparse_geodesic_fusion (human-origin)

```python
def ei_sparse_geodesic_fusion(
    component_scores: Iterable[np.ndarray],
    nonnegative_weights: np.ndarray,
    calibrator: tuple[float, float],
) -> np.ndarray:
```

ei_sparse_geodesic_fusion (generated-evolved-origin)

```python
    """Generated/evolved EI: sparse nonnegative fusion plus geodesic odds."""
    weights = np.asarray(nonnegative_weights, dtype=np.float64)
    weights = np.maximum(weights, 0.0)
    weights = weights / (float(np.sum(weights)) + 1.0e-12)

    stacked = np.vstack([np.asarray(score, dtype=np.float64) for score in component_scores])
    fused = np.sum(weights[:, None] * stacked, axis=0)
    fused = np.clip(fused, 1.0e-6, 1.0 - 1.0e-6)

    slope, intercept = calibrator
```

ei_sparse_geodesic_fusion (mixed-origin)

```python
    logit = np.log(fused / (1.0 - fused))
    calibrated = 1.0 / (1.0 + np.exp(-np.clip(slope * logit + intercept, -40.0, 40.0)))
    return calibrated
```

References

[1] M. I. Skolnik, *Radar Handbook*, 3rd ed. New York, NY, USA: McGraw-Hill, 2008.

[2] M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.
[3] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-19, no. 4, pp. 608–621, Jul. 1983.
[4] V. C. Chen, *The Micro-Doppler Effect in Radar*. Norwood, MA, USA: Artech House, 2011.
[5] D. Tahmoush, "Review of micro-doppler signatures," *IET Radar, Sonar & Navigation*, vol. 9, no. 9, pp. 1140–1146, 2015.
[6] D. L. Hall and J. Llinas, "An introduction to multisensor data fusion," *Proceedings of the IEEE*, vol. 85, no. 1, pp. 6–23, 1997.
[7] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. New York, NY, USA: Wiley, 2001.
[8] A. Karlsson, "Radar measurements on drones, birds and humans with a 77GHz FMCW sensor," Zenodo, 2021. [Online]. Available: https://zenodo.org/records/5896641
[9] D. Gusland, J. M. Christiansen, B. Torvik, F. Fioranelli, S. Z. Gurbuz, and M. Ritchie, "Open radar initiative: Large scale dataset for benchmarking of micro-doppler recognition algorithms," in *2021 IEEE Radar Conference (RadarConf21)*. IEEE, 2021.
[10] I. Roldan, C. R. del Blanco, A. Duque Quevedo, F. Ibanez Urzaiz, J. Gismero Menoyo, A. Asensio Lopez, D. Berjon, F. Jaureguizar, and N. Garcia, "Dopplernet: A convolutional neural network for recognising targets in real scenarios using a persistent range-doppler radar," *IET Radar, Sonar & Navigation*, vol. 14, no. 4, pp. 593–600, 2020.
[11] S. Rahman and D. A. Robertson, "Radar micro-doppler signatures of drones and birds at K-band and W-band," *Scientific Reports*, vol. 8, no. 1, p. 17396, 2018.
[12] P. Molchanov, R. I. A. Harmanny, J. J. M. de Wit, K. Egiazarian, and J. Astola, "Classification of small UAVs and birds by micro-doppler signatures," *International Journal of Microwave and Wireless Technologies*, vol. 6, no. 3–4, pp. 435–444, 2014.
[13] M. Ritchie, F. Fioranelli, H. Borrion, and H. Griffiths, "Multistatic micro-doppler radar feature extraction for classification of unloaded/loaded micro-drones," *IET Radar, Sonar & Navigation*, vol. 11, no. 1, pp. 116–124, 2017.